Guidance and Control of Unmanned Autonomous Vehicles

(Course Code: AE 700)

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Practice Set 3

1. Consider a fixed-wing Micro Air Vehicle (MAV) navigating a straight-line path, denoted as $\mathcal{P}_{line}(r, q)$. The path originates at the waypoint $w_1 = (100, 200, -150)$ m and is directed toward a terminal waypoint located at $w_2 = (900, 800, -150)$ m. The MAV's current position is given by $p = (400, 600, -150)$ m.

   (a) Compute the course angle of the path, χ_q, in degrees.

   (b) Determine the lateral cross-track error, e_{py}, relative to the desired path.

   Ans: $\chi_q = 36.87°$, $e_{py} = 140\ m$.

2. An MAV is executing a longitudinal flight maneuver along an inclined three-dimensional straight-line path $\mathcal{P}_{line}(r, q)$. The path is defined by an origin vector $r = (0, 0, -200)$ m and a unit direction vector $q = (0.8, 0, 0.6)^\top$. The vehicle is currently situated at position $p = (500, 100, -400)$ m. Determine the desired altitude, h_d, shown in Figure 1, corresponding to the MAV's current position.

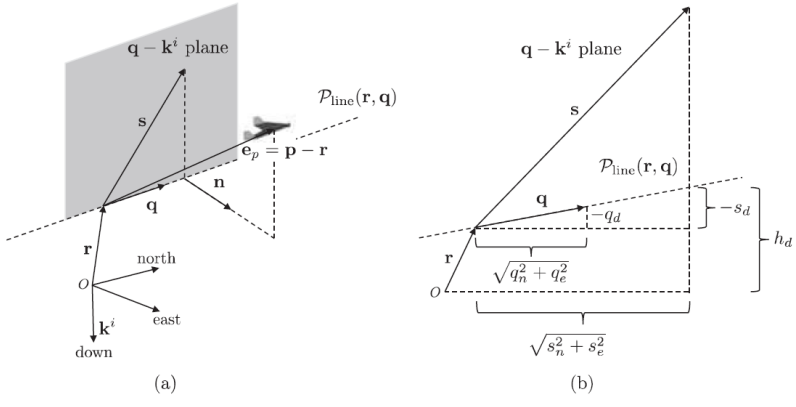

Figure 1: Desired altitude calculation for straight-line path following in longitudinal direction.

   Ans: $h_d = -175\ m$

3. The UAV is assigned to follow a straight-line nominal flight path defined by a starting waypoint $\mathbf{w}_1$ and an ending waypoint $\mathbf{w}_2$. The coordinates of the waypoints are given as:$\mathbf{w}_1 = \begin{bmatrix} 0 & 0 & 0 \end{bmatrix}^\top$ and $\mathbf{w}_2 = \begin{bmatrix} 3 & 4 & 0 \end{bmatrix}^\top$. Find out the shortest distance of the UAV from the line joining the waypoints when the UAV is located at

(a) $\mathbf{p}_A = \begin{bmatrix} 3 \\ 0 \\ 5 \end{bmatrix}$ (b) $\mathbf{p}_B = \begin{bmatrix} -3 \\ -4 \\ 5 \end{bmatrix}$ (c) $\mathbf{p}_C = \begin{bmatrix} 6 \\ 8 \\ 10 \end{bmatrix}$

Ans: (a) 5.5 units (b) 5 units (c). 10 units

4. Suppose we want to construct a path that a vehicle can follow, with a minimum turn radius of $R = 500$. We have been given a few waypoints [specified by $\boldsymbol{w}_1 = (0, 0, 1000)^\top$, $\boldsymbol{w}_2 = (1000, 1000, 2000)^\top$, and $\boldsymbol{w}_3 = (0, 2000, 1000)^\top$] to consider for planning such a path. Owing to the vehicle's turn rate constraints, the path designer used a suitable fillet to connect the straight-line paths between waypoints as shown in Figure 2.

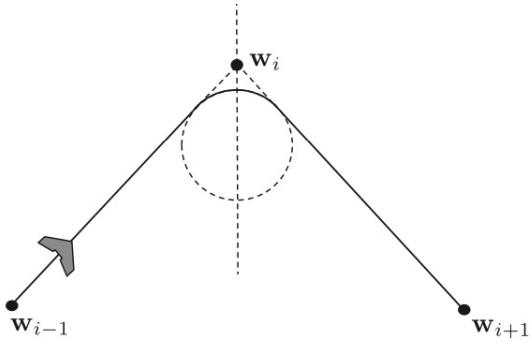

Figure 2: The transition from straight line path between the waypoints.

(a) Find out the center of the fillet along with the locations at which the curved path starts and ends.

(b) Compute the change in path length for a curved path compared to when the vehicle follows straight-line paths.

Ans: (a) $\boldsymbol{c} = [387.6803 \quad 1000 \quad 1387.6803]^\top$, $\boldsymbol{r}_1 = [591.7517 \quad 1408.2483 \quad 1591.7517]^\top$, $\boldsymbol{r}_2 = [591.7517 \quad 591.7517 \quad 1591.7517]^\top$
(b) -458.8834 m

5. Consider a rigid UAV with an RGB-D sensor (RGB and depth) mounted in a gimbal for navigation. Assume the inertial and body frame coincide, hence $\phi = \theta = \psi = 0°$. The camera points at an object at an azimuth angle of 45° and an elevation angle of −30°. Determine the global coordinates of the object whose pixel coordinates in the image frame are $(10, 20)$, with a range of 2 m, obtained from the depth sensor. (**Camera parameters:** Horizontal field of view - 90°, image size (in terms of pixels) - 640(width) x 480(height).)

Ans: $p^i_{obj} \approx \begin{bmatrix} 1.13 & 1.22 & 1.11 \end{bmatrix}^\top$ *m*

6. Obtain the gimbal's desired azimuth and elevation angles so that it can point towards the object at $(10, 5, -3)^\top$ in the inertial frame. The position of MAV w.r.t the inertial frame is estimated as $(2, -3, -5)^\top$, and the Euler angles corresponding to the standard rotation of the inertial frame to the body frame are given by $\phi = 30°, \theta = 45°, \psi = 60°$.

Ans: $\alpha^c_{el} = 54.7369°, \alpha^c_{az} = 17.8655°$

7. Compute Dubins path length, given the position p, the course χ, and the radius R. The objective of Dubin's path is to have an optimal transition from one configuration to another. The start configuration is $p_s = (0, 0)$ and $\chi_s = \frac{\pi}{2}$. The end configuration is $p_e = (10, 10)$ and $\chi_e = 0$. The radius $R = 2$.

   Ans: 14.45 units

8. A MAV is flying at an altitude of 150 m above a perfectly flat plain. The camera identifies a target, and the normalized line-of-sight vector computed in the inertial frame is $\check{\boldsymbol{l}}^i = (0.5, 0.5, 0.707)^\top$. Estimate the range to the target $\mathbb{L}$.

   Ans: 212.16 m

9. A MAV is flying straight and level ($\phi = 0°, \theta = 0°, \psi = 0°$) at an altitude of 100 m above a flat earth. The gimbal is pointing exactly forward and down, with $\alpha_{az} = 0°$ and $\alpha_{el} = -45°$. The target is perfectly centered in the image plane ($\epsilon_x = 0, \epsilon_y = 0$).

   (a) Calculate the nominal North position of the target relative to the MAV.

   (b) The flat-earth geolocation equation is highly sensitive to attitude estimation errors. If the onboard attitude estimator has a bias of $+2°$ in pitch ($\hat{\theta} = 2°$), what is the resulting error in the estimated North position of the target?

   Ans: 100 m and 7.2 m

10. An approaching target has an apparent size of 50 pixels in the image plane ($\epsilon_s = 50$). By differentiating sequential frames, you determine that the target size is growing at a rate of 5 pixels/second ($\dot{\epsilon}_s = 5$). Assume the target is located directly along the optical axis, meaning the pixel coordinates are $\epsilon_x = \epsilon_y = \mathbf{0}$. If the focal length is 400 pixels, calculate the time to collision t_c.

    Ans: 10 s

11. A MAV is in a climbing right turn. Its onboard rate gyros measure body rates of $p = 0.1$ rad/s, $q = 0.05$ rad/s, and $r = 0.2$ rad/s. The pan-tilt gimbal is actively tracking a stationary ground target, with current gimbal angles $\alpha_{az} = 30°$ and $\alpha_{el} = -45°$, and gimbal rates $\dot{\alpha}_{az} = -0.1$ rad/s and $\dot{\alpha}_{el} = 0.05$ rad/s. The target is currently located at pixel coordinates $\epsilon_x = 100$, $\epsilon_y = -50$. The camera has a focal length $f = 500$ pixels. Calculate the apparent angular motion vector $\dot{\check{\boldsymbol{l}}}^c_{\text{app}}$ of the target in the camera frame induced solely by the rotation of the MAV and gimbal.

    Ans: $\dot{\check{\boldsymbol{l}}}^c_{\text{app}} = \frac{1}{512.35}\begin{bmatrix} 11.5 \\ -6.5 \\ -2.95 \end{bmatrix} = \begin{bmatrix} 0.0224 \\ -0.0127 \\ -0.0058 \end{bmatrix} \; s^{-1}$